

OOS Impact Analytics — Methodology Sheet

How the dashboard detects out-of-stock and demand-throttling events and values their revenue / contribution-margin impact. Worked example: **VV-VITA-208 — Horse Chestnut Capsules Organic** (amazon.de). Prepared for review.

1. Data sources

- **Amazon FBA Inventory Ledger** — real daily sellable warehouse balance and units shipped per SKU. The account runs **Pan-EU**, so stock is pooled: a SKU is physically out only when the whole EU balance hits zero.
- **Novadata margin export** — daily units, sales and contribution margin (CM3) per SKU and marketplace; supplies price and CM3 per unit for valuation.
- **Shopify catalog** — English product titles.

2. Two core measures

Demand rate (lambda). Trailing 90-day average units per calendar day, per SKU and marketplace. Daily sales follow an approximate Poisson process, so the chance of selling zero on a normal day is $P(0)=e^{-\lambda}$: ~37% at 1/day, ~5% at 3/day, ~0.7% at 5/day. A zero-sales day is therefore only treated as a stock-out when lambda is high enough that zero is a real anomaly (default threshold 3/day).

Reach (days-of-supply). EU sellable stock divided by trailing 28-day average units shipped — how many days of cover remain.

3. Daily classification (one label per SKU x marketplace x day)

Evaluated in priority order; the first match wins:

Priority	Label	Condition	Counts as
1	Physical OOS	EU sellable balance = 0	Lost (involuntary)
2	Critically low	Reach < 3 days	Lost (involuntary)
3	Cooling down	3 <= reach < 30 days AND throttled (price >= +8% and/or ad spend cut >= 50%)	Miss (voluntary)
4	Marketplace gap	Units = 0 on a day enclosed by sales, lambda >= 3/day	Lost (involuntary)

Guards: the SKU must already be selling (ignores pre-launch); a zero-run must be enclosed by later sales or be within the last 21 days (ignores discontinued tails). Cooling-down separates deliberate throttling — which otherwise looks like a demand drop — from genuine stock-outs, so the two are never double-counted.

4. Impact valuation

On any flagged day: **missed units = max(lambda - actual units, 0)** (the expected demand we didn't capture). Valued at the SKU's trailing average selling price and CM3 per unit:

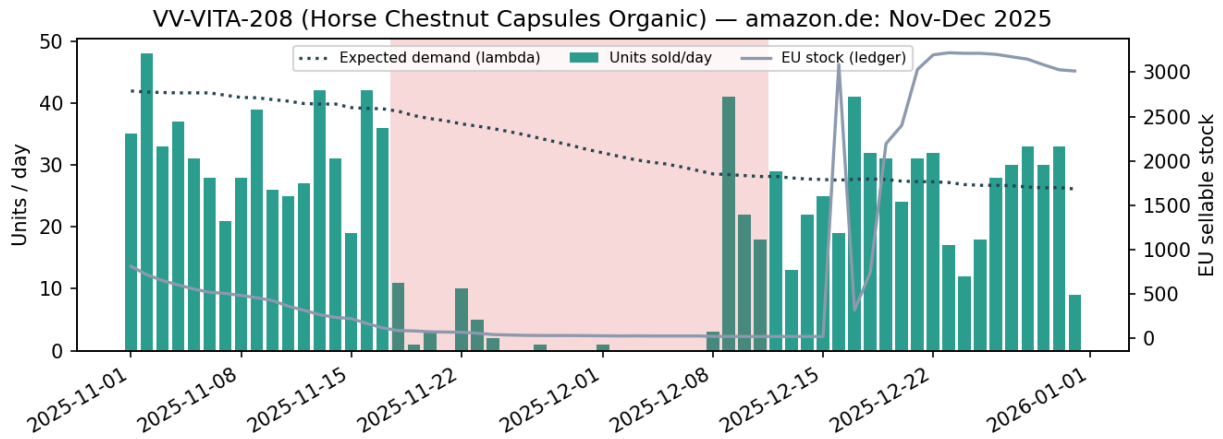
Lost / Miss revenue = missed units x avg price

Lost / Miss CM3 = missed units x avg CM3 per unit (the true P&L impact).

'Lost' = involuntary (OOS); 'Miss' = voluntary (cooling down). Consecutive flagged days are grouped into discrete events.

5. Worked example

VV-VITA-208 — Horse Chestnut Capsules Organic, amazon.de. Full year on this marketplace: **24 OOS days** (lost €13.435 revenue / €2.741 CM3) and **4 cooling-down days** (miss €485 revenue / €82 CM3).



Nov–Dec 2025: EU stock fell from ~800 to ~30 units for a SKU selling 30–50/day (reach collapsed). Sales flatlined for ~2 weeks until the 16 Dec restock. The EU balance never hit literally zero (floor ~23), so the **reach < 3** rule — not a balance-of-zero rule — is what flags this real lost-sales episode.

Representative days (Lost/Miss shown as CM3 €):

Date	Units	λ	EU stk	Reach	Price	Category	Lost €	Miss €
11-16	42	39	170	4	19.91	—		
11-18	11	39	90	2	20.12	Critically low (<3d)	111	
11-19	1	38	86	2	19.90	Critically low (<3d)	148	
11-20	3	37	76	2	19.90	Critically low (<3d)	138	
11-21	0	37	73	2	—	Critically low (<3d)	149	
11-22	10	37	71	2	20.33	Critically low (<3d)	107	
11-23	5	36	61	2	19.90	Critically low (<3d)	125	
11-24	2	36	45	1	19.90	Critically low (<3d)	135	
11-25	0	35	40	1	—	Critically low (<3d)	141	
11-26	0	35	36	1	—	Critically low (<3d)	139	
11-27	1	34	34	1	19.90	Critically low (<3d)	132	
11-28	0	34	34	1	—	Critically low (<3d)	134	
11-29	0	33	34	1	—	Critically low (<3d)	132	
11-30	0	32	32	1	—	Critically low (<3d)	129	
12-01	1	32	31	1	19.90	Critically low (<3d)	122	
12-02	0	31	29	1	—	Critically low (<3d)	124	
12-03	0	31	30	2	—	Critically low (<3d)	122	
12-04	0	30	29	2	—	Critically low (<3d)	123	
12-05	0	30	29	2	—	Critically low (<3d)	122	
12-06	0	30	29	2	—	Critically low (<3d)	121	
12-07	0	29	29	2	—	Critically low (<3d)	118	
12-08	3	29	24	2	19.90	Critically low (<3d)	103	

6. Caveats

Impact figures are estimates (assume the SKU would have sold at its recent rate). Ad-spend-cut detection only applies from when Novadata began reporting Advertising Costs (~Feb 2026); the price-hike lever spans the full year. Reach is EU-pooled, so reach-based rules need ledger coverage for the SKU. All thresholds (3/day, reach 3 & 30, +8%, -50%) are tunable in the dashboard.